Chapter 5. Responsive features, the learner, and the population: efficient simulation of language contact dynamics

Henri Kauhanen

5 May 2025

5.1 Introduction

Chapter REF defined a *responsive feature* as any linguistic feature which is sensitive to the acquisition setting. In particular, L2-difficult features are responsive in this sense: by definition, such features pose a difficulty for, and only for, acquirers beyond a critical age. The Trudgill conjecture (Chapter REF) predicts that such features are prone to being lost if enough acquirers beyond this critical age are found in the speech community. Writing on simplification of adjective inflection in Bergen Norwegian, Trudgill proposes:

> My suggestion is that when the proportion of non-native becomes as close to 50% as this – the proportion of non-native speakers of Norwegian in Bergen was certainly very high (4,000 out of 9,000) – the number of face-to-face dialect-contact type interactions, and therefore potential instances of accommodation [...], would have reached a threshold level at which some aspects of the non-native variety could transfer to the native (Trudgill 2011, 57–58).

The present chapter undertakes a quantitative and computational recapitulation of this proposal, asking just how many post-critical-age acquirers must be present in the speech community for L2-difficult features to be lost. If there indeed exists a threshold of simplification due to the responsive nature of some or other feature, what is this threshold? Fifty percent L2 learners? Sixty percent? Or possibly a number whose precise value depends, in some way, on other factors involved in the contact situation? Moreover, how long must the contact situation continue for simplification to be permanent, so that the L2-difficult feature is not simply restored once L2 learners are removed from the population? Finally, how do sociodemographic factors such as the frequency of interactions between L2 learners and L1 speakers affect the likelihood of simplification?

The challenges involved in answering these questions ought to be clear from the outset. An empirical answer would require, at a minimum, a sample of language histories some of which culminate in loss of an L2-difficult feature, with others culminating in its retention, together with reliable estimates of the likely numbers of post-critical-age acquirers in each case, combined with information about further factors such as the degree of L2-difficulty exhibited by the feature in question, the duration for which L2 learning was in place, and the frequencies of different patterns of interaction between the different subpopulations. It is clear that empirical data are not available to this extent, at this resolution, in most cases of interest. As a consequence, a mixed methodology, combining empirical data, theory, and modelling, is called for.

The strategy adopted in the present chapter is to draw upon the predictive power of mechanistic modelling (Geritz and Kisdi 2012). It is first demonstrated how feature responsivity can be captured in a simple stochastic model of language acquisition. A mixed population is then introduced, one in which some learners acquire the language before the critical age, the remaining being post-critical-age acquirers; for ease of exposition, I will refer to the former as L1 learners, and the latter as L2 learners, in what follows.[1] As these learners interact, the frequency of the L2-difficult feature may go up or down in the population; we are chiefly interested in the question under what circumstances this frequency vanishes to zero, corresponding to full simplification.

The chapter builds upon work published in Kauhanen (2022). In that article, the above questions were answered in a particular limiting case of the model, known as the deterministic limit. This special case is characterized by the following idealizing assumptions: (a) there are infinitely many learners in each group (L1 and L2), (b) learners pick interlocutors entirely at random, and (c) all learners within a group behave the same. For completeness, the results of the earlier paper are restated and discussed below (in Section 5.3). However, the present chapter ventures beyond the deterministic limit, exploring a number of empirically relevant scenarios of the full stochastic model in computer simulations. A methodological innovation – efficient simulation of large populations through the use of a so-called Beta approximation to describe the statistical equilibrium of language learning – is also introduced and discussed.

FIXME: add some shpiel about the results of the stochastic simulations

5.2 Responsive features in language acquisition

Throughout this chapter, it is assumed that language learners need to decide between two grammars, one of which carries a responsive feature. I will denote the L2-difficult grammar with G_1 and its competitor (which is assumed not to involve any L2-difficulty) with G_2.[2] The grammars accept languages L_1 and L_2, respectively, understood as sets of well-formed strings, constructed out of a common alphabet. In

[1] It is important to bear in mind that, with this nomenclature, our L2 learners do not comprise individuals who exhibit childhood bilingualism – such individuals learn an L2 but learn it before the critical age and hence the responsive features do not respond in their case. See Chapter REF for more on this point.

[2] These assumptions are made in order to keep the model simple, without losing the essential intuition of competition between a responsive grammar and a non-responsive one. It would be relatively straightforward to generalize the model for multiple competing grammars, each of which may involve some (potentially non-zero) amount of L2-difficulty. This and other potential avenues for further research are discussed in more length in Section 5.8.

general, L_1 and L_2 may overlap; cf. Figure 1. The symbol a_1 will be used to denote the probability of someone employing G_1 producing a string belonging to the set $L_1 \setminus L_2$ (i.e. a string that G_2 does not accept), and similarly a_2 will be used for the probability of someone employing G_2 producing a string in $L_2 \setminus L_1$. These quantities will be referred to as the *advantages* of the two grammars in what follows. They are also assumed to reflect the weak generative capacities of the two grammars (cf. Yang 2000) and may thus be assumed to be constant within any given competition situation.

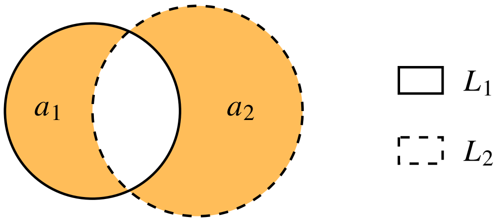

Figure 1. Two grammars, G_1 and G_2, generate languages L_1 and L_2, represented here as intersecting sets. The shaded areas are the sets $L_1 \setminus L_2$ and $L_2 \setminus L_1$; their sizes represent the probabilities that strings from these sets are produced, a_1 and a_2 (referred to as the advantages of G_1 and G_2, respectively).

Each language user is assumed to embody a pair of *weights* for the two grammars; formally, these weights constitute a categorical probability distribution over the grammars. The weight of G_1 will be denoted P and the weight of G_2 with $Q = 1 - P$; these are random variables whose values change in response to the linguistic input received by the language user or learner. These changes could in principle be implemented by any number of imaginable mechanisms, and it is one of the goals of language acquisition research to disentangle probable from improbable mechanisms of learning. In order to obtain a reasonably tractable mathematical model, I follow the variational learning framework of Yang (2002), itself founded upon the pioneering work of Bush and Mosteller (1955) on general statistical mechanisms of reinforcement.

Let us first outline how the learning algorithm works for an acquirer under the critical age. Upon receiving an input sequence $s \in L_1 \cup L_2$, the learner chooses one of the two grammars to use: G_1 with probability P, and G_2 with probability Q. The learner then attempts to parse the input sequence. If the chosen grammar accepts the input, this grammar's weight is increased and the weight of its competitor decreased. If the grammar does not accept the input, the reverse operation is carried out, decreasing this grammar's weight and increasing the competitor's weight.

Concretely, the changes made to the weights follow the linear reward–penalty scheme of Bush and Mosteller (1955). The mathematics of this are summarized in Table 1, off of which the updates to P and Q can be read under all possible circumstances. The number γ is a learning rate parameter which regulates the magnitude of the updates the learner makes to the weight distribution; this parameter must satisfy the technical requirement $0 < \gamma < 1$.[3]

Table 1. Learning algorithm for L1 learner (Bush and Mosteller 1955; Yang 2002). The new values of P and Q are $P' = P + \Delta P$ and $Q' = Q + \Delta Q$. A positive ΔP (resp. ΔQ) rewards G_1 (resp. G_2), while a negative value punishes it.

input string	grammar chosen	update to P	update to Q
$s \in L_1 \setminus L_2$	G_1	$\Delta P = \gamma Q$	$\Delta Q = -\gamma Q$
$s \in L_1 \setminus L_2$	G_2	$\Delta P = \gamma Q$	$\Delta Q = -\gamma Q$
$s \in L_1 \cap L_2$	G_1	$\Delta P = \gamma Q$	$\Delta Q = -\gamma Q$
$s \in L_1 \cap L_2$	G_2	$\Delta P = -\gamma P$	$\Delta Q = \gamma P$
$s \in L_2 \setminus L_1$	G_1	$\Delta P = -\gamma P$	$\Delta Q = \gamma P$
$s \in L_2 \setminus L_1$	G_2	$\Delta P = -\gamma P$	$\Delta Q = \gamma P$

To model feature responsivity, Kauhanen (2022) introduced a simple modification to this basic algorithm. For a learner past the critical age, a second learning rate parameter δ is introduced whose effect is to render the acquisition of G_1 more difficult for the learner.[4] The update rule is given in Table 2. It is obvious that under this modification of the algorithm, grammar G_1 faces an inherent disadvantage compared to grammar G_2: even when the former is rewarded after parsing success, an amount of weight is deducted (specifically, that amount is δP), whereas no such deduction occurs for G_2 (by contrast, to retain the identity $P + Q = 1$, the amount δP is always *added* to the weight of G_2).

For reasons which will become obvious shortly, it is useful to define the quantity

$$d = \delta / \gamma,$$

i.e. the ratio of the two learning rate parameters. Note that for a pre-critical-age learner $d = 0$, whereas for a post-critical-age learner $d > 0$. The ratio d expresses the degree of L2-difficulty of G_1, normalized by the general learning rate γ. It is thus a general measure of L2-difficulty whose value does not depend on the overall speed with which the learner adjusts the grammar weights P and Q.

Table 2. Learning algorithm for L2 learner (Kauhanen 2022). The new values of P and Q are $P' = P + \Delta P$ and $Q' = Q + \Delta Q$.

input string	grammar chosen	update to P	update to Q
$s \in L_1 \setminus L_2$	G_1	$\Delta P = \gamma Q - \delta P$	$\Delta Q = -\gamma Q + \delta P$
$s \in L_1 \setminus L_2$	G_2	$\Delta P = \gamma Q - \delta P$	$\Delta Q = -\gamma Q + \delta P$
$s \in L_1 \cap L_2$	G_1	$\Delta P = \gamma Q - \delta P$	$\Delta Q = -\gamma Q + \delta P$
$s \in L_1 \cap L_2$	G_2	$\Delta P = -\gamma P - \delta P$	$\Delta Q = \gamma P + \delta P$
$s \in L_2 \setminus L_1$	G_1	$\Delta P = -\gamma P - \delta P$	$\Delta Q = \gamma P + \delta P$

[3]Technically, $\gamma = 1$ is also possible, leading to a model in which learners "hop" categorically between G_1 and G_2 (cf. Gibson and Wexler 1994; Niyogi and Berwick 1996). I leave this special case aside here.

[4]To guarantee that P and Q always remain probabilities, a technical requirement is that $0 < \delta < 1 - \gamma$.

input string	grammar chosen	update to P	update to Q
$s \in L_2 \setminus L_1$	G_2	$\Delta P = -\gamma P - \delta P$	$\Delta Q = \gamma P + \delta P$

Let c_1 denote the probability with which the learner receives an input sequence which G_1 does not accept, and let c_2 similarly denote the probability with which an input sequence not accepted by G_2 is received. These "penalty probabilities" (Yang 2000) determine the ultimate course of learning in the following sense. If it is possible to assume that the values of c_1 and c_2 do not change during the learning period (i.e. the penalty probabilities are constant), we say the learner operates in a *stationary* learning environment. If, moreover, both c_1 and c_2 are non-zero, the learner's weights for G_1 and G_2 converge onto a stationary distribution with increasing learning iteration (Bush and Mosteller 1955; Narendra and Thathachar 1989). In practice this means that, under this learning algorithm, any learning trajectory can be broken down into two phases, a transient phase during which the values of P and Q are moving towards stationarity, and a stationary phase in which statistical equilibrium has been reached. This is illustrated in Figure 2.

Since $P+Q=1$, we may concentrate on P, the weight on the L2-difficult grammar G_1, in what follows. The mean and variance of P at stationarity may be expressed in closed form. Let $\mathrm{E}[P]$ stand for the mean (i.e. expected value) and $\mathrm{Var}[P]$ for the variance of P. At equilibrium,

$$\mathrm{E}[P] = \frac{c_2}{c_1 + c_2 + d} \tag{1}$$

and

$$\mathrm{Var}[P] = -\mathrm{E}[P]^2 + \frac{CG^2 + 2c_2G\gamma}{1 - G^2 + 2CG\gamma}\mathrm{E}[P] + \frac{c_2\gamma^2}{1 - G^2 + 2CG\gamma} \tag{2}$$

with $C = 1 - c_1 - c_2$ and $G = 1 - \gamma - \delta$ (see Kauhanen 2022, Supplementary file, equations (20) and (24)).

A few remarks are in order. First, once statistical equilibrium has been reached, the expected value of the learner's weight only depends on two things: (i) the penalty probabilities c_1 and c_2, and (ii) the L2-difficulty factor d. In particular, this expected value is not affected by the absolute magnitudes of the learning rate parameters. The variance, on the other hand, is: the higher the learning rates, the higher the variance, and consequently the more fluctuations we observe at equilibrium (Figure 2). This will play a crucial role later when we break away from the deterministic limit.

Secondly, the expected weight on G_1 (the grammar which contains the L2-difficult feature) at equilibrium is always lower for an L2 learner compared to an L1 learner, and the difference is greater the higher the degree of L2-difficulty d is.

Thirdly, although we have not proved it here, the approach to stationarity is exponential. This fact (which is also apparent from the simulated trajectories in Figure 2) is important for the following reason: we need not in general worry about whether learners reach equilibrium; convergence is quick. A related fact (again not strictly proved here) is that the learner's initial state does not influence the stationary distribution; the learner is "ergodic" (Narendra and Thathachar 1989).

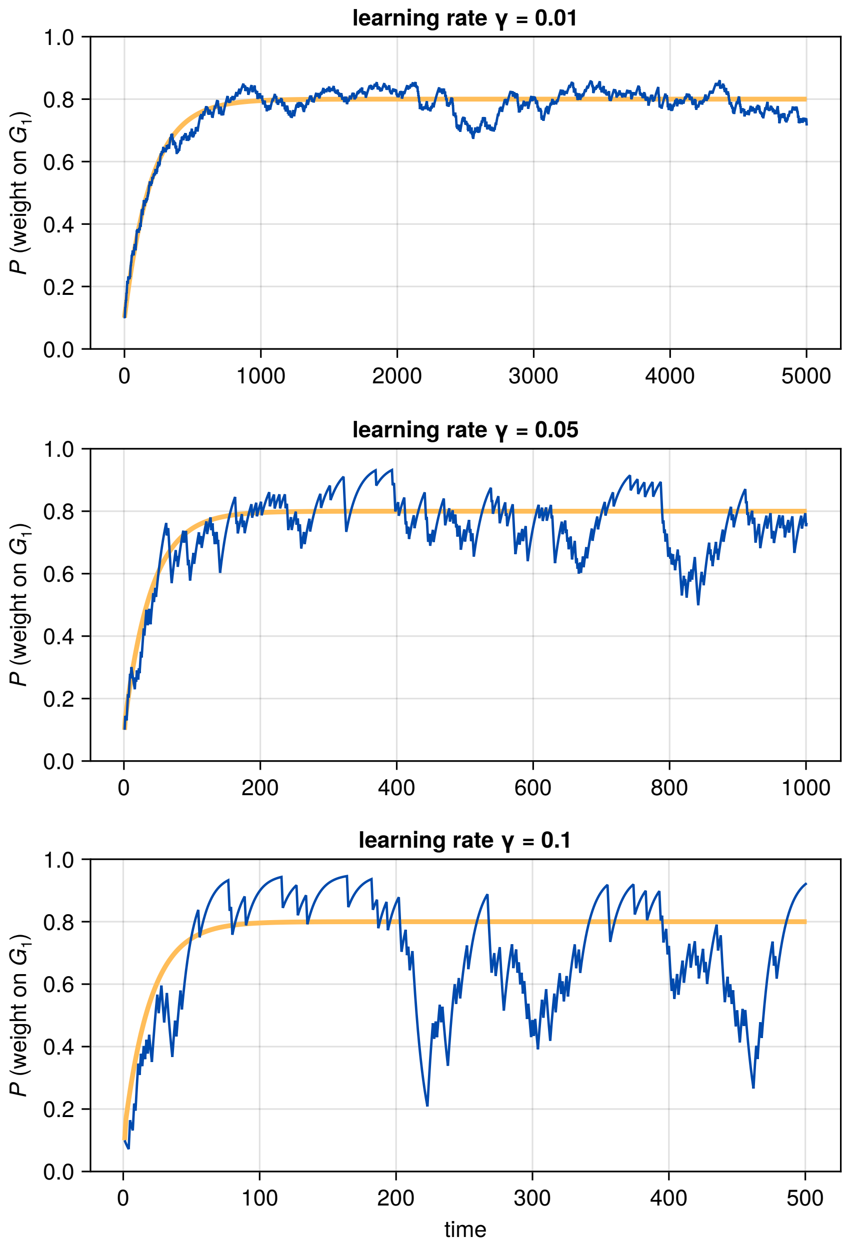

Figure 2. Three simulated learners, each in the same learning environment characterized by penalty probabilities $c_1 = 0.05$ and $c_2 = 0.4$ but with different learning rates: $\gamma = 0.01$, $\gamma = 0.05$ and $\gamma = 0.1$. Each learner additionally is subject to an L2-difficulty of $d = 0.05$. With these choices, the expected value of P at stationarity is 0.8 (the thick yellow curve gives the evolution of the expected value). With the parameters chosen, and with each learner starting from $P(0) = 0.1$, the stationary distribution is reached after about the first $10\gamma^{-1}$ learning iterations.

Most work on the variational learning model has pretended that the variance does not exist at equilibrium (e.g. Yang 2000; Heycock and Wallenberg 2013; Simonenko, Crabbé, and Prévost 2019; Kauhanen 2022). This allows one to assume that every learner in a population of learners is described by a single number, namely the expectation $\mathrm{E}[P]$. It is then relatively straightforward to write down an inter-generational equation (in either discrete or continuous time) which describes how the weight in one generation of learners is determined from the value of that weight in an immediately preceding generation of learners (see Wallenberg et al. REF). Section 5.3 reviews this strategy particularly as it arises in the modelling of contact-induced change. Section 5.4 reintroduces the variance; the remaining sections then ask to what extent the predictions made by the deterministic limit continue to hold when populations are not deterministic.

5.3 The deterministic limit and the bifurcation threshold

In real populations, the penalty probabilities c_1 and c_2 of the competing grammars may be different from learner to learner (depending on which other language users they interact with) and need not remain constant over the duration of learning. Analysing a model that attempts to reproduce this reality incurs significant challenges – the numerous interdependencies between different language users in such a model mean that analytical results are (nearly) unobtainable. It makes sense to start from a simpler vantage point, one which employs a small set of simplifying assumptions that together render the model analytically tractable. This is the approach adopted in Kauhanen (2022), briefly summarized next.

Assume a mixed population of L1 and L2 learners. Every learner internalizes weights P and Q for two grammars, as outlined above. Let x refer to the probability of encountering an L1 learner employing G_1 at a given point in time (mathematically, this is equivalent to the expected value of P, aggregated over the L1 population). Similarly, let y refer to the probability of encountering an L2 learner employing G_1. The state of the population is described by the pair of numbers (x, y); since these are probabilities, the state space is the unit square $[0,1] \times [0,1]$. We now ask: what is the eventual fate of the dynamics? Does the state (x, y) tend to some definite value, and if so, what is that value? Moreover, under what conditions do we have $(x, y) \to (0,0)$, so that no one (the L1 learners included) uses the L2-difficult grammar in the end?

Suppose we make the following assumptions:

1. Learners are arranged in a discrete sequence of non-overlapping generations.
2. Each generation consists of infinitely many learners.
3. Each learner in generation n picks interlocutors from generation $n-1$ uniformly at random.
4. There is zero variance at any learner's stationary distribution.

Under these assumptions, Kauhanen (2022) shows that (x, y) tends to a global attractor $(\hat{x}, \hat{y})$ which either is the origin $(\hat{x}, \hat{y}) = (0,0)$ or lies in the interior of the unit square. In the former case G_1, the grammar with the responsive feature, eventually gives way completely to grammar G_2, which incurs no L2-difficulty. It is important to note that, in this case, the effects of L2-difficulty also percolate to the L1 learner population, even though the latter themselves face no L2-difficulty in learning: the presence of L2 learners shifts the linguistic input L1 learners are exposed to, lowering the probability of them encountering strings which are uniquely accepted by the L2-difficult grammar G_1. Through this mechanism, the effects of L2 learning may percolate through the whole population – at least in principle.

Whether this sort of "total percolation" or "total simplification" (total replacement of G_1 by G_2) occurs, depends on whether σ, the proportion of L2 learners in the population, exceeds a critical threshold σ_{crit} whose value is

$$\sigma_{\text{crit}} = \left(1 - \frac{a_2}{a_1}\right)\left(1 + \frac{a_2}{d}\right) \tag{3}$$

(Kauhanen 2022, equation (19)). Note that the critical threshold is a relation involving four parameters: σ, the proportion of L2 learners; d, the amount of L2-difficulty experienced by the L2 learners; and a_1 and a_2, the advantages of the competing grammars.

Equation 3 can be used to make and test predictions about the outcomes of individual language contact situations. One of the empirical case studies considered by Kauhanen (2022) concerns the expression of null subjects in Afro-Peruvian Spanish (APS), a variety of Spanish spoken by descendants of people brought as slaves to Colonial Peru in the 17th–19th centuries. Sessarego (2015) argues that modern-day APS displays a mixed system of null subjects, analysable as grammar competition between a consistent null subject grammar and a non-null subject grammar. Null subjects, on the other hand, have been found to incur L2-difficulty in a number of experimental studies (Bini 1993; Pérez-Leroux and Glass 1999; Margaza and Bel 2006). From the point of view of sociolinguistic typology, then, the challenge in this particular empirical case study is to explain why, despite obvious extensive linguistic admixture in Colonial Peru, APS did not fully lose null subjects.

Using available demographic data (which is admittedly scant), Kauhanen (2022) estimated the likely proportion of L2 learners at around $\sigma \approx 0.5$, while corpus estimates by Simonenko, Crabbé, and Prévost (2019) put the advantage of the null subject grammar G_1 at $a_1 = 0.7$ and the advantage of the competing non-null subject grammar G_2 at $a_2 = 0.05$. Although we at present lack any empirical estimates of the magnitude of the L2-difficulty parameter d in this (or indeed in any other) case, the critical value σ_{crit} has a lower bound which is attained as $d \to \infty$; this is

$$\lim_{d \to \infty} \sigma_{\text{crit}} = 1 - \frac{a_2}{a_1}.$$

Plugging in the advantage parameters, we find

$$1 - \frac{a_2}{a_1} = 1 - \frac{0.05}{0.7} \approx 0.93.$$

Since obviously $0.5 < 0.93$, the prediction that APS settles on a mixed system is thus borne out.

5.4 Embracing variation

Adopting the simplifying assumptions listed in the previous section made it possible to derive Equation 3, the bifurcation threshold for contact-induced simplification in the deterministic limit. In relation to real-world application, this result suffers from the following main limitations, which arise immediately from the simplifying assumptions made:

1. It assumes *complete and random mixing*. Everybody can come into contact with everybody else; in particular, there is no sense in which L2 learners might be more likely to interact among each other than with L1 learners (as might have been realistically the case in, for instance, the sorts of colonial settings which gave rise to languages such as APS).
2. It does not allow for *variation between learners in the same (L1/L2) group*. Yet the L2 acquisition literature abounds in empirical data which strongly suggests that some individuals attain higher fluency than others, and even the usage patterns of L1 speakers vary from person to person.
3. It assumes a *constant proportion of L2 learners* σ. However, in several empirically interesting situations (including the history of APS), it is the case that σ first goes up, then comes down again as the children of immigrants acquire the language as an L1 rather than as an L2.

Relaxing these assumptions requires us to view the system as an agent-based model and to run computer simulations, something that introduces a number of technical challenges. For instance, even though it was pointed out above that the approach to a learner's stationary distribution is exponential, this may still take thousands of learning iteratations depending on the specific values of the learning rate parameters and the penalty probabilities. If there are thousands of learners in the population, the numbers quickly multiply, leading to long simulation runtimes. While this is often not a problem for running a single simulation, a single simulation outcome is never sufficient to give anything approaching a complete picture of the model's possible behaviours. First, to understand the role played by stochastic factors, the outcomes of multiple simulation runs must be summarized (typically by taking averages). Secondly, and more importantly, the many model parameters involved result in a combinatorial explosion if we wish to sweep over the parameter space in systematic fashion.

To illustrate this, suppose it took one minute to run a single simulation of interest – a reasonable estimate for a reasonably complex model run on a modern computer. If there are three model parameters, each of which can take on 10 different values, then running the simulation for all possible model parameter combinations will take 10^3, i.e. one thousand, minutes. This corresponds to about 16 hours, still a reasonable time to wait perhaps. But now note that, as emphasized above, we may wish to repeat the simulation for each model parameter combination a number of times, say 50 times, to be able to assess the effect of stochastic factors. Unless it is possible to parallelize the simulation code, this immediately raises the runtime to 830 hours, or just over one month. Adding just one more model parameter to the sweep (assuming again 10 different possible values for the parameter) multiplies the runtime by 10, requiring us to wait 10 months for our results. Requiring more resolution of the parameter sweep (e.g. 100 possible values for each parameter) is clearly impossible in practice, unless we have access to a supercomputer.

It is here that the availability of some analytical results becomes very useful. As explained in Section 5.4, in a stationary learning environment the behaviour of learners may be summarized by the properties of the distribution of P at statistical equilibrium. Although the exact distribution is unknown, its mean and variance can be computed, as noted in Section 5.2. It turns out that these are enough to characterize that distribution in practice.

Specifically, we adopt the following strategy – which could be termed the "Beta approximation" – in the following simulations. Instead of simulating each learner's learning trajectory up to statistical equilibrium, we sample the learner's ultimate value of P from a Beta distribution with suitable shape parameters α and β.[5] The shape parameters are obtained via a method known as moment matching. The mean and variance of the Beta distribution are

$$\mu = \frac{\alpha}{\alpha+\beta}$$

and

$$\sigma^2 = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

respectively (see e.g. REF). Thus, to obtain a Beta distribution with a desired mean and variance, we simply need to solve the above equations for the shape parameters α and β, and plug in the mean and variance. But since we know the mean and variance of the learner's stationary distribution, we may now use these numbers to obtain a Beta distribution that approximates that stationary distribution.

To illustrate, Figure 3 shows the results of a number of experiments; model parameters were drawn at random and are listed in Table 3. In each experiment, 10,000 learners were simulated up to statistical equilibrium. The histograms depict that equilibrium, i.e. the distribution of P across all 10,000 learners.[6] The curves give the probability density of the Beta distribution found through the method of moment matching. It is evident that, in each case (i.e. for different values of the model parameters c_1, c_2, γ and δ), the agreement is very good, justifying the use of the Beta approximation.

[5] The Beta distribution is a continuous probability distribution over the interval $[0,1]$ and is thus a suitable choice to describe the distribution of a random variable such as our P. It is parameterized by two parameters, $\alpha > 0$ and $\beta > 0$, which determine the distribution's shape.

[6] In fact, since the learning scheme is ergodic and each learner within a given experiment faces the same learning environment (the same penalty probabilities c_1 and c_2) and employs the same learning rate parameters, the same result could be obtained by sampling 10,000 values from the stationary distribution of a single learner.

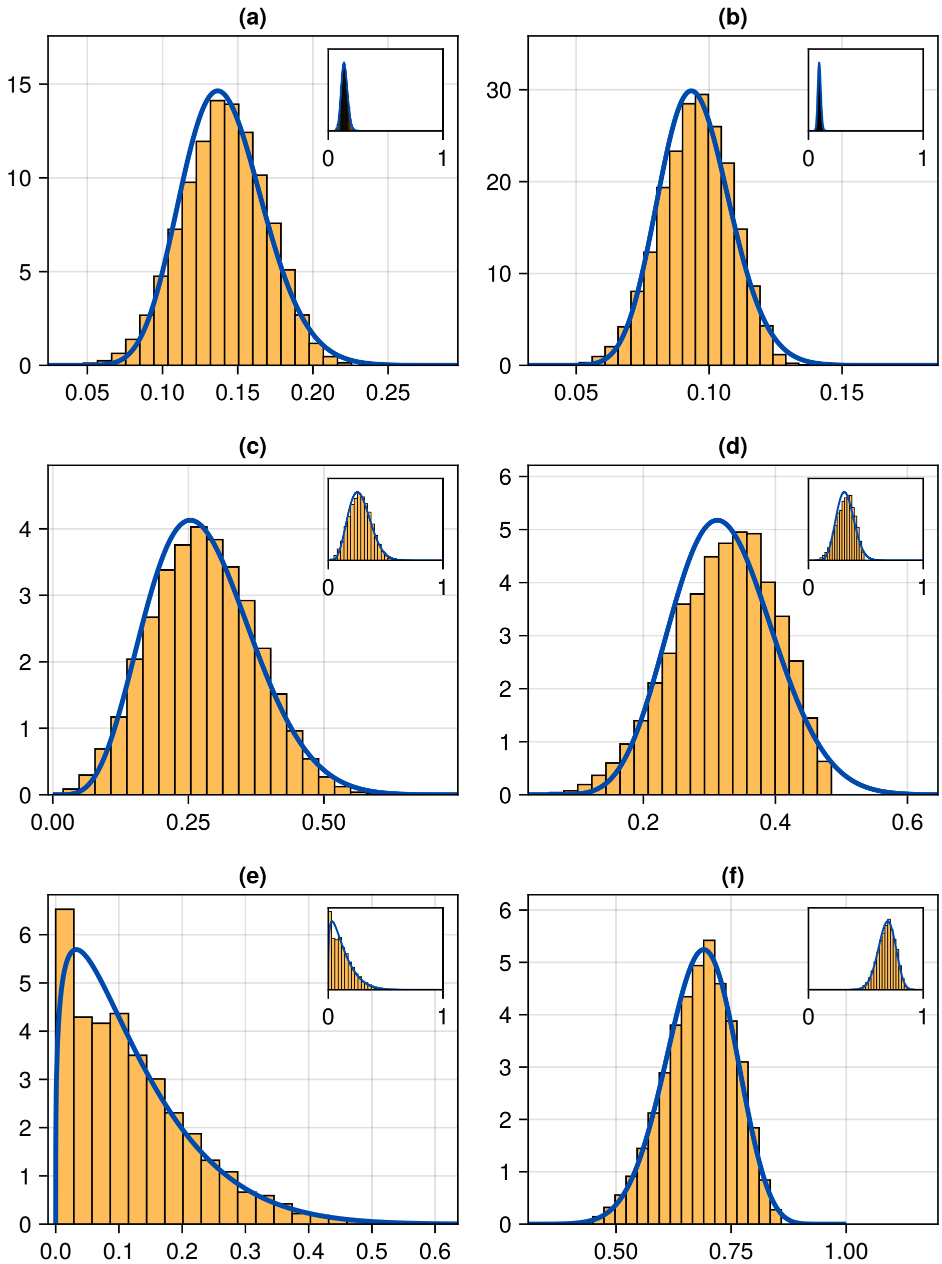

Figure 3. Six learning simulation experiments. In each experiment, 10,000 learners were simulated to equilibrium with randomly drawn parameters (see Table 3). The equilibrium distribution of P, the weight on G_1 (yellow histogram), is well approximated by a Beta distribution (blue curves indicating the probability density function) found by moment matching. Insets scale each facet to the entire range of possible values of P, the interval $[0, 1]$.

Table 3. Model parameters (c_1, c_2, γ and δ; randomly drawn) for the simulation experiments of Figure 3, together with the mean (E[P]) and variance (Var[P]) of the stationary distribution (Equation 1 and Equation 2) and the moment-matched shape parameters (α, β) of the approximating Beta distribution.

	c_1	c_2	γ	δ	E[P]	Var[P]	α	β
(a)	0.94	0.66	0.027	0.083	0.14	0.0008	22.6	137.6
(b)	0.87	0.73	0.012	0.073	0.1	0.0002	45.3	431.1
(c)	0.34	0.31	0.081	0.038	0.28	0.0091	5.8	15.2
(d)	0.14	0.57	0.085	0.09	0.32	0.0058	11.8	24.9
(e)	0.32	0.09	0.1	0.033	0.12	0.0093	1.3	9.2
(f)	0.25	0.87	0.082	0.013	0.68	0.0056	25.5	12.0

As a consequence, simulation runtimes are drastically reduced. On the computer used to simulate the above experiments,[7] the median runtime for simulating one learner to equilibrium was 468 µs. By contrast, sampling from the Beta distribution only took 342 ns on average, meaning that the latter operation is about one thousand times (three orders of magnitude) faster.

FIXME: the above runtime numbers are from my laptop, aren't they? Need to move the experiment code to the part of the codebase that is run on the simulation computer!

5.5 Simulation setup

For the agent-based simulations, the following setup is assumed. Simulations are carried out on a spatial substrate which is a regular two-dimensional lattice of $i \times j$ sites with non-periodic boundaries, i giving the horizontal dimension and j the vertical dimension. While L1 learning may occur in any part of this space, L2 learning is constrained to occur only in the lower half of the lattice (i.e. in sites with a vertical index less than half of j). The shape of the lattice is varied through the use of an *aspect ratio* parameter ρ, defined as the ratio of lattice width to lattice height. For $\rho > 1$, the lattice is wider than it is tall, and for $\rho < 1$, it is taller than it is wide (see Figure 4).[8]

The decision to constrain L2 learning to the lower half of the lattice, and the use of the aspect ratio parameter ρ, together allow us to control the intensity of the initial interactions between L1 and L2 learners, i.e. interactions between the top and bottom halves of the lattice. This structure may be interpreted either as spatial (as in many situations of language contact between two contiguous geographical regions) or as social (as in situations of colonization and slavery) in origin.

[7] A PC with an Intel Core i5-10600K processor running at 5 GHz and 32 GB of DDR4 RAM at 4000 MHz. All simulation code was written in Julia (Bezanson et al. 2017), version 1.11.4, and can be obtained from https://github.com/erc-starfish/FIXME.

[8] Concretely, the horizontal and vertical dimensions are set as $i = \lfloor \sqrt{\rho L} \rfloor$ and $j = \lfloor \sqrt{\rho^{-1} L} \rfloor$ where L is the approximate total number of sites on the lattice.

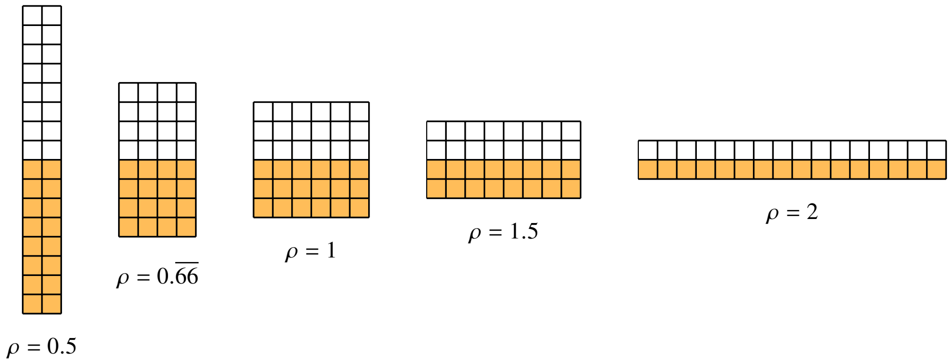

Figure 4. The aspect ratio parameter ρ is used to set the relative horizontal and vertical dimensions of the lattice. L2 learning is constrained to occur only in the bottom half of the lattice (yellow), and so ρ implicitly controls the length of the interface between the initial L1 and L2 learning regions. (Lattices in the actual simulations are far larger, with a total of $L \approx 5000$ sites.)

Simulations proceed in discrete time. At each time step, three actions are performed:

1. Birth and L1 acquisition. A number of speakers are created and made to undergo the process of L1 acquisition. The number of speakers "born" this way is proportional to (i) a birth rate parameter, β, (ii) the current population size, N, and (iii) a carrying capacity, K; the latter ensures that population sizes do not explode exponentially (which would lead to computationally intractable, and also empirically unrealistic, simulation runs). Concretely, the number of newborn speakers is sampled from a binomial distribution with success probability β and number of trials $\lfloor N(1-(N/K)) \rfloor^{+}$, where $\lfloor x \rfloor^{+} = \lfloor x \rfloor$ (i.e. the integer part of x) if $x > 0$ and $\lfloor x \rfloor^{+} = 0$ if $x < 0$. Each newborn speaker is assigned to a lattice site at most R sites away from the site occupied by a randomly chosen already existing speaker. Multiple speakers may occupy the same site; once placed on the lattice, a speaker stays in the same site for its lifetime. L1 acquisition is modelled by sampling from a Beta distribution as explained in Section 5.4. The learner estimates the penalty probabilities c_1 and c_2 from its local environment. Concretely, the learner samples F speakers at random from within a distance (Chebyshev distance) of R on the lattice, or takes all neighbouring speakers if their number does not exceed F. The local penalty probabilities c_1 and c_2 are then calculated from the frequencies of use of grammars G_1 and G_2 among these neighbours, together with the advantage parameters a_1 and a_2, in the usual manner: if x stands for the local frequency of G_1, then $c_1 = a_2(1-x)$ and $c_2 = a_1 x$ (see e.g. Yang 2000).

2. Immigration and L2 acquisition. A number of speakers are "imported" into the lattice and made to undergo the process of L2 acquisition. The number of immigrants is, like the number of newborn agents in step 1, dependent on both the current population size and the carrying capacity, and is similarly sampled from a binomial distribution, except that a migration rate parameter τ is used instead of the birth rate β. This way of arranging the immigration process is modelled on the assumptions that (i) larger populations attract more immigration, and (ii) a finite substrate can support only a finite amount of immigration. L2 acquisition proceeds as L1 acquisition, via the Beta approximation, except that the learner is also subject to the L2-difficulty parameter d, set as a global parameter for all learners.

In the Appendix, it is shown that given the above general assumptions about population dynamics, the expected proportion of L2 learners in the population is

$$\mathrm{E}[\sigma] = \frac{\tau}{\beta + \tau}$$

once the populations have stabilized. In other words, this quantity may be readily computed from the birth and immigration rates β and τ. Conversely, in order to obtain a desired expected proportion of L2 speakers $\mathrm{E}[\sigma]$ having set some fixed birth rate β, we may invert the above equation to verify that the immigration rate τ needs to be chosen so that

$$\tau = \frac{\mathrm{E}[\sigma]}{1 - \mathrm{E}[\sigma]}\beta. \tag{4}$$

In the following simulations, this is what we do: i.e. rather than directly set the immigration rate τ, we set the desired proportion of L2 learners σ and calculate τ using Equation 4.

3. Death. When agents are introduced to the population, they are assigned an age which is 0 for L1 learners and a random number for L2 learners; this random number is constructed by drawing a real number between 0 and an upper limit λ from a truncated normal distribution with mean m and standard deviation s, and rounding down to an integer. Agents are also assigned a lifetime, sampled from a Weibull distribution with shape parameter k and scale parameter λ. At each simulation iteration, the age of every agent on the lattice is incremented by one, and agents whose ages exceed their lifetimes are removed from the lattice. The particular choices of k and λ used in the simulations (see Table 4) imply that the survival distribution is left-skewed, meaning that death is more likely the older a speaker becomes.

[FIXME: maybe put a figure which has distributions: the number of births, the number of immigrants, the age of an immigrant, lifetime]

Each simulation is initialized with 100 L1 speakers with full use of G_1 ($P = 1$) inserted in random locations of the L1 portion of the lattice. From here, the simulation proceeds according to the above description, with the exception that for the first 500 simulation steps, there is no immigration. This is to ascertain that full use of G_1 is in fact a stable population equilibrium in the absence of L2 learning. After the first 500 steps, immigration is enabled and continues for the following T_{L2} steps. This parameter allows us to control the temporal extent of the immigration process, facilitating the modelling of both situations of temporally extended moderate bilingualism and more radical population dynamics such as those of colonial societies, which are characterized by strong initial immigration followed by its disappearance at abolition. Each simulation is run for a total of T time steps. The main observable of interest is the population average of P, i.e. the mean usage of the L2-difficult grammar G_1, throughout and, in particular, at the end of the simulation.

It is important to emphasize that while L2 acquisition is constrained to only occur in the lower half of the lattice, no such constraint is in place for L1 acquisition. Thus, when L2 learners have children (i.e. when newborn speakers are placed in the vicinity of existing speakers in the lower half of the lattice), those children undergo their acquisition process as L1 learners.

All model parameters are summarized in Table 4. Parameters which were held constant across all simulations are indicated with those constant values. Parameters with values labelled 'various' were varied across simulations; further information on these will be given below.

Table 4. Simulation parameters.

Parameter	Symbol	Value(s)
Advantage of grammar G_1	a_1	various
Advantage of grammar G_2	a_2	various
Learning rate	γ	various
L2-difficulty of G_1	d	various
Lattice aspect ratio	ρ	various
Approximate lattice size (number of sites)	L	5000
Interaction radius	R	10
Maximum interaction density (number of contacts)	F	50
Birth rate	β	0.01
Expected proportion of L2 learners	$\mathrm{E}[\sigma]$	various
Length of immigration period	T_{L2}	various
Mean L2 learner age at immigration	m	200
S.D. of L2 learner age at immigration	s	100
Number of L1 speakers at time 0	N_0	100
Shape parameter of death distribution	k	10.0
Scale parameter of death distribution	λ	1000.0
Population carrying capacity	K	various
Length of simulation (steps)	T	various

Figure 5 illustrates a typical simulation run, plotting the mean weight on G_1, the proportion of L2 learners, and normalized population size (the number of speakers divided by its maximum value over the duration of the simulation) against time, for the simulation parameters listed in Table 5. As L2 learners are introduced into the population from time 500 onwards, we observe a steady decrease in the mean weight on G_1. Concomitantly, the proportion of L2 learners increases with the influx of these learners. Ultimately, the weight on G_1 tends toward zero; this aligns with the prediction of the deterministic model, as the proportion of L2 learners exceeds the critical thredshold σ_{crit} given the parameter values of Table 5. The L2-difficult grammar is not recovered, at least not instantly, even when the supply of L2 learners is cut off later.

Table 5. Parameter values for the simulation run of Figure 5 and Figure 6. For the values of the constant parameters, refer to Table 4.

Parameter	Symbol	Value(s)
Advantage of grammar G_1	a_1	0.12
Advantage of grammar G_2	a_2	0.1
Learning rate	γ	0.01
L2-difficulty of G_1	d	10
Lattice aspect ratio	ρ	1.3
Expected proportion of L2 learners	$\mathrm{E}[\sigma]$	0.7
Length of immigration period	T_{L2}	4000
Population carrying capacity	K	1000
Length of simulation (steps)	T	10000

Figure 6 provides an alternative view of the same simulation run. The six panels of this figure provide a snapshot of the lattice at six different time points, with crosses indicating speakers (upright crosses = L1 learners, slanted crosses = L2 learners; blue/dark = higher weight on G_1, pink/light = lower weight on G_1). To promote legibility, a random sample of 20% of all speakers is drawn, and at slightly jittered locations.

5.6 Systematic parameter sweeps

In all, the simulation setup described in the previous section involves 18 model parameters. If one were to carry out a systematic parameter sweep across this parameter space, with, say, 10 values per parameter and 50 repetitions for each parameter combination, the number of simulation runs would equal $50 \cdot 10^{18}$. This is intractable even with the Beta approximation. Some insight is thus required in identifying what parameters are likely to be of interest. This, in turn, requires us to formulate specific research questions.

In what follows, I will focus on the following two questions:

1. Does the bifurcation threshold identified in the deterministic limit, i.e. Equation 3, retain its validity in this stochastic simulation setup?
2. How do language contact outcomes depend on (i) interaction patterns between the subpopulations (i.e. on lattice aspect ratio ρ) and (ii) the length of the period of language contact (i.e. the length of immigration parameter T_{L2})?

To probe question 1, a sweep was carried out by varying the parameters σ (or $\mathrm{E}[\sigma]$), the (expected) proportion of L2 learners, and d, the extent of L2-difficulty on grammar G_1 (see Table 6 for model parameters). For each combination of σ and d, 50 simulations were carried out, each simulation lasting 10,000 time steps. As explained

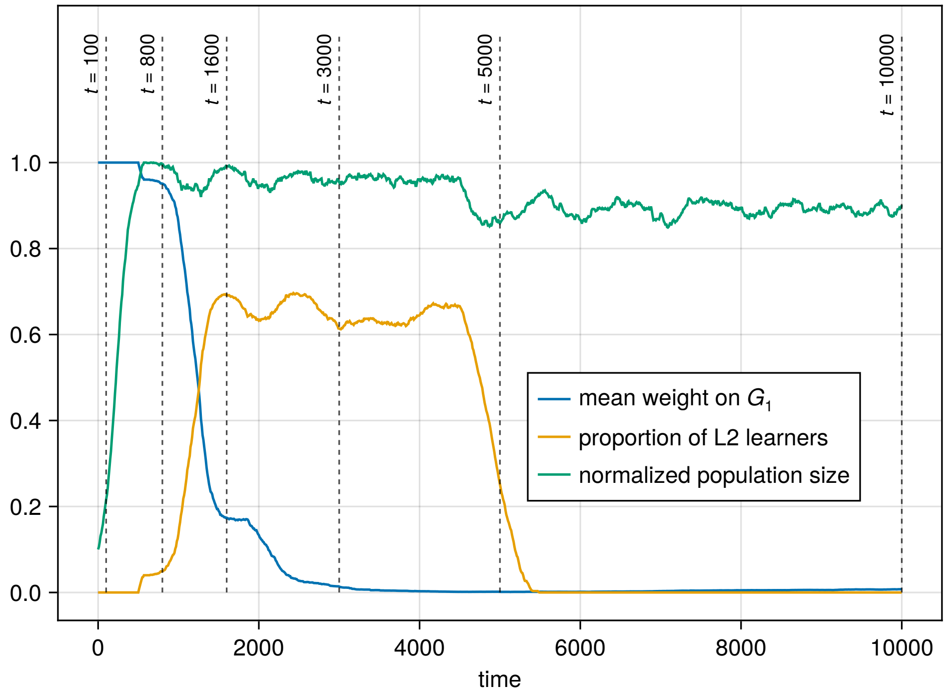

Figure 5. One simulation run. For model parameters, see Table 5. Snapshots of the population at the indicated time points are provided in Figure 6.

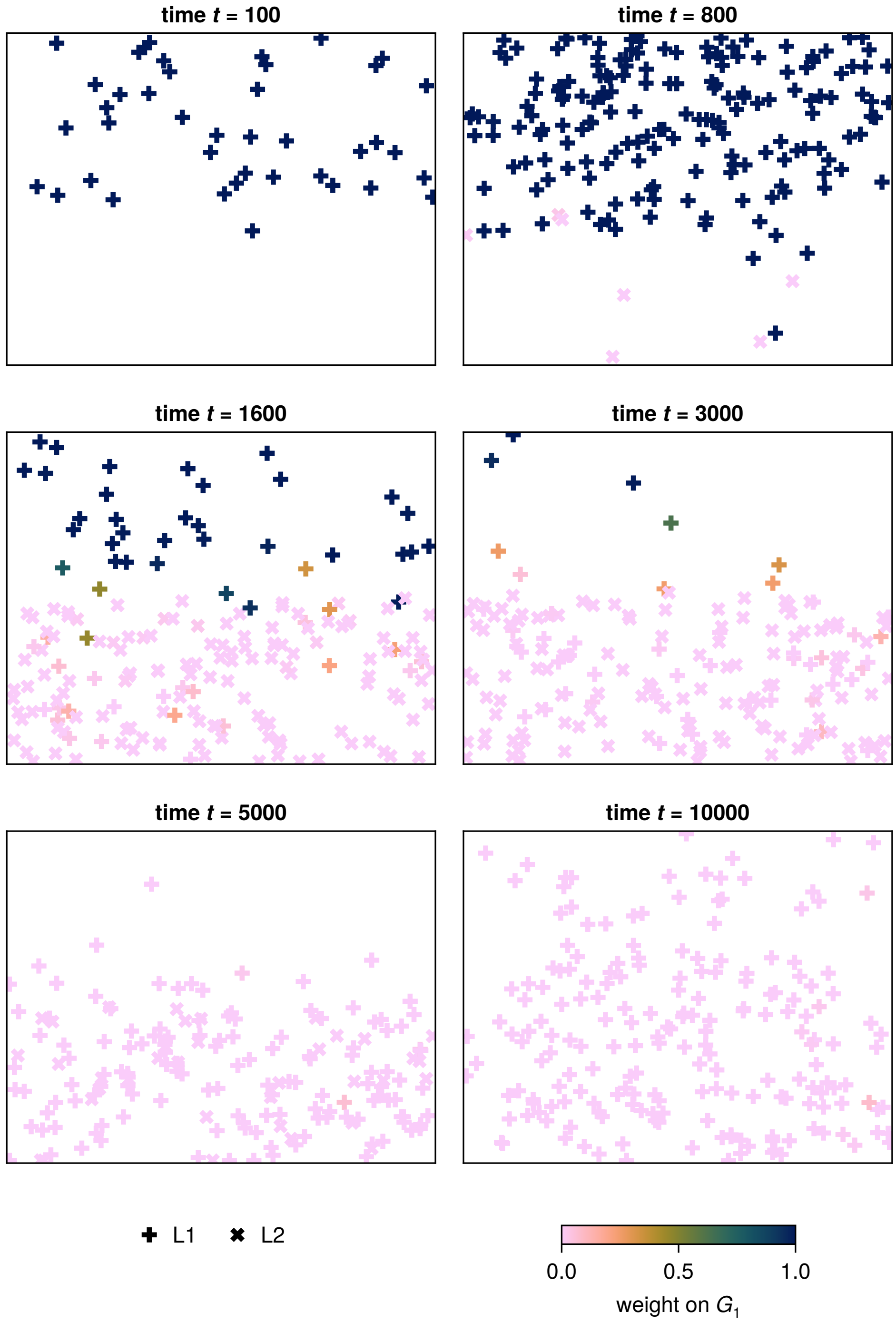

Figure 6. Snapshots from a simulation with the parameters of Table 5. Each cross represents one speaker in its position in the lattice (a random sample of 20% of all speakers is actually visualized, and speaker positions are slightly jittered, for reasons of legibility). Upright crosses represent L1 learners, slanted crosses L2 learners. Colour corresponds to the speaker's weight on the L2-difficult grammar G_1.

above, L2 learning was enabled from time step 500; moreover, in this particular sweep, the supply of L2 learners was never cut off.

For each combination of proportion of L2 learners σ and L2-difficulty d (and advantage parameters a_1 and a_2; see Table 6), Equation 3 predicts a specific threshold σ_{crit}. This threshold is depicted in Figure 7 as the thick dashed line. The heatmap provides the eventual value of P (weight on G_1), averaged over the population and over all 50 simulation runs. Furthermore, contour lines are drawn for P across this heatmap. We find that the stochastic simulations are in general agreement with the theoretical (deterministic-limit) prediction of Equation 3. Interestingly, however, the theoretical calculation appears to somewhat overestimate the extent of simplification: the contour line closest to the theoretically predicted σ_{crit} is 0.1, corresponding to 10% use of G_1 across the entire population.[9]

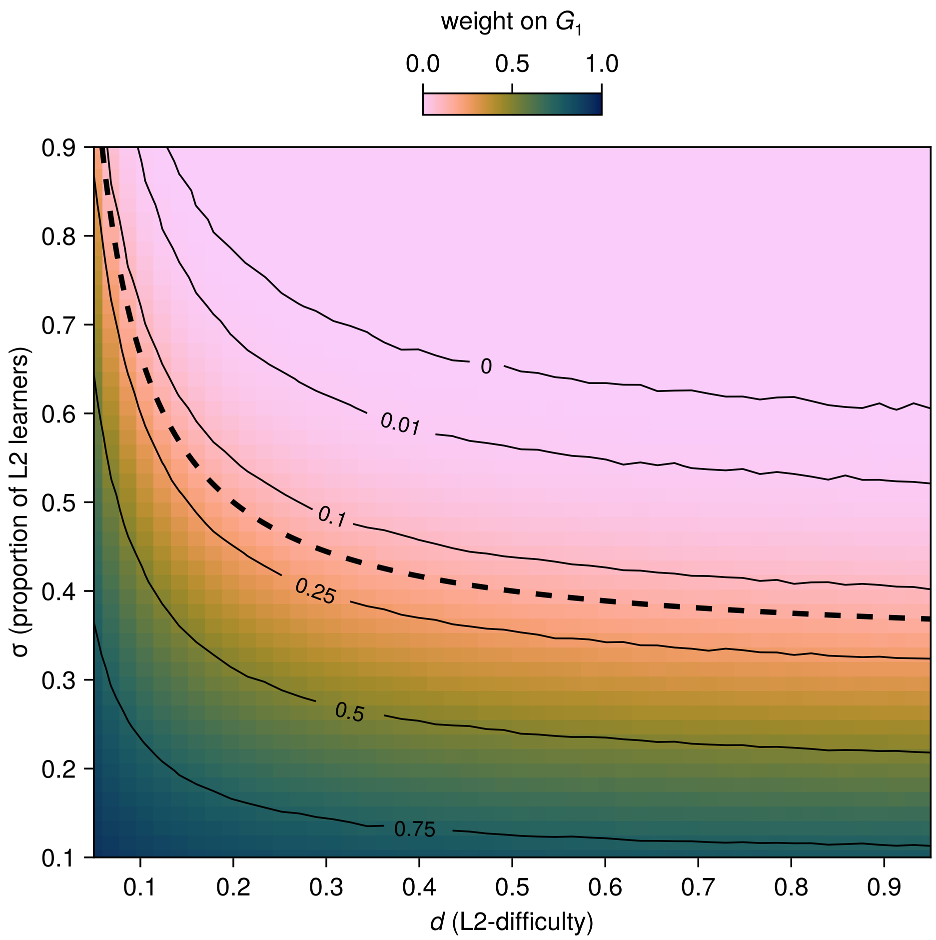

Figure 7

[9]This is not due to imperfect convergence of the simulation runs. The same experiment was repeated for 30,000 steps instead of the 10,000 of the original experiment, with essentially the same outcome.

Table 6. Parameter values for the sweep of Figure 7. In this and subsequent tables, the notation $[a \overset{n}{\ldots} b]$ refers to a sequence of n equally spaced values from a to b, endpoints inclusive; $\log_{10}[a \overset{n}{\ldots} b]$ is the same except with logarithmic spacing. For the values of the constant parameters, refer to Table 4.

Parameter	Symbol	Value(s)
Advantage of grammar G_1	a_1	0.15
Advantage of grammar G_2	a_2	0.1
Learning rate	γ	0.01
L2-difficulty of G_1	d	$[0.05 \overset{50}{\ldots} 0.95]$
Lattice aspect ratio	ρ	1.0
Expected proportion of L2 learners	$\mathrm{E}[\sigma]$	$[0.1 \overset{50}{\ldots} 0.9]$
Length of immigration period	T_{L2}	10000
Population carrying capacity	K	1000
Length of simulation (steps)	T	10000

Turning to question 2, a second sweep was performed for varying lattice aspect ratios σ and varying durations of the L2 learning period T_{L2}; for all model parameters, see Table 7). Again, 50 simulations were carried out for every simulation combination, this time for 30,000 time steps. Figure 8 visualizes the results, with time step on the horizontal axis, aspect ratio ρ on the vertical axis, and T_{L2} varying across the panels. We observe an interaction between ρ and T_{L2}: for short periods T_{L2}, tall lattices ($\rho < 1$) conserve the simplification longer than wide lattices ($\rho > 1$). With higher T_{L2}, the effect effectively disappears.

This interaction with short periods of L2 learning may seem counterintuitive at first: wide lattices ($\rho > 1$) are precisely those that have a longer active interface between the initial L1 and L2 subpopulations. One might thus expect such lattices to favour simplification. However, the active interface works in both directions, i.e. a longer interface also promotes the (anti-simplificatory) effect that L1 speakers have on L2 learners. A taller lattice, by contrast, gives rise to an extended domain in the lower half of the lattice in which simplification can proceed to completion, with the children of the L2 learners acquiring the simplified grammar as L1.

Table 7. Parameter values for the sweep of Figure 8. For the values of the constant parameters, refer to Table 4.

Parameter	Symbol	Value(s)
Advantage of grammar G_1	a_1	0.12
Advantage of grammar G_2	a_2	0.1
Learning rate	γ	0.01
L2-difficulty of G_1	d	10
Lattice aspect ratio	ρ	$\log_{10}[0.001 \overset{31}{\ldots} 1000.0]$
Expected proportion of L2 learners	$\mathrm{E}[\sigma]$	0.7
Length of immigration period	T_{L2}	$[1000 \overset{6}{\ldots} 11000]$

Parameter	Symbol	Value(s)
Population carrying capacity	K	1000
Length of simulation (steps)	T	30000

5.7 Afrikaans and Afro-Peruvian Spanish revisited

To further test the model, we can ask to what extent its specific predictions on the two case studies considered in Kauhanen (2022) hold, in ~~the~~ light of what is known empirically in both cases. The two case studies are null subjects in Afro-Peruvian Spanish (already discussed above) and verbal inflection in Afrikaans. These two cases not only correspond to somewhat different sociohistorical circumstances, they also involve different linguistic properties and different outcomes. As discussed above, the move from null subjects in the direction of overt subjects in APS is only partial; present-day APS has not simplified the L2-difficult null subject grammar completely, and younger generations of speakers of APS in fact use null subjects more than older generations (Sessarego 2014). Verbal inflection in Afrikaans, however, was completely simplified, in the sense that all regular verbs only possess one syncretic form across the paradigm of person and number (for details, see Ponelis 1993). A successful computational model of these dynamics must therefore be able to predict both developments: the "failure" to simplify in one case, and the "success" in the other.

Such empirical evaluation is tricky, however, due to the many parameters involved and the difficulty in their estimation. Several of the sociodynamic parameters are simply unknown, and there is no conceivable way of retrieving the required information from an empirical source (such is the case of, for example, the mean age of L2 learners at immigration, m; cf. Table 4). Such parameters can, however, be estimated (with some uncertainty) from the available historiographic data, which pertains to known numbers of Europeans and non-Europeans both in the Dutch Cape Colony and in Colonial Peru at the relevant time periods. Table 8 and Table 9 give these numbers, slightly simplified from the tables in Kauhanen (2022), which in turn were obtained from the demographic studies in Giliomee and Elphick (1979) and Bowser (1974).

Table 8. Demographic data for Afrikaans. From Giliomee and Elphick (1979) by way of Kauhanen (2022).

Year	Europeans	Non-Europeans
1670	125	65
1690	788	429
1711	1693	1834
1730	2540	4258
1750	4511	5676
1770	7736	8572
1798	20000	27454
1820	42975	33711

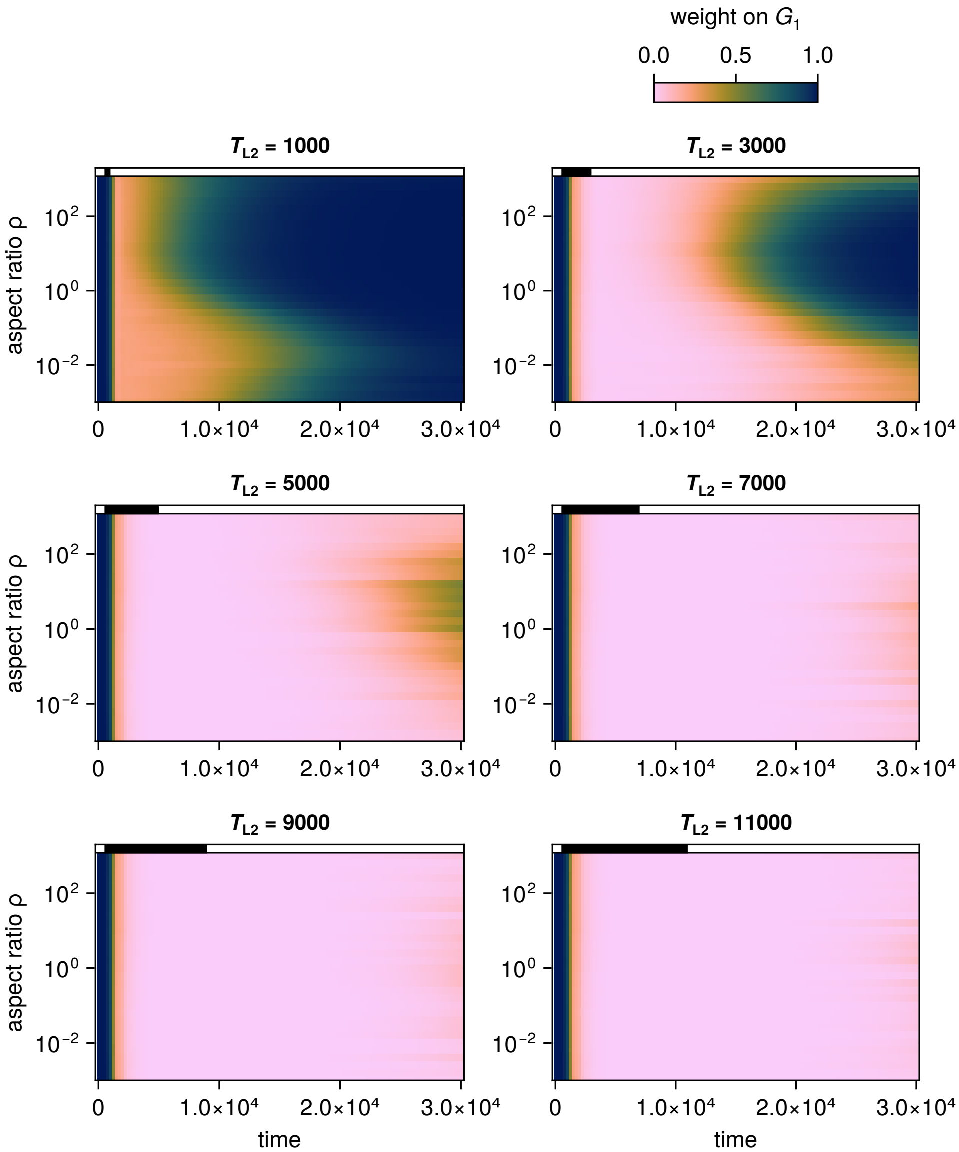

Figure 8. Results of a parameter sweep crossing lattice aspect ratio (ρ) and the length of the period of L2 learning/immigration (T_{L2}). The horizontal axis in each facet represents simulation time step; the period of L2 learning is depicted by the black bar at the top of each panel.

Table 9. Demographic data for Afro-Peruvian Spanish. From Bowser (1974) by way of Kauhanen (2022).

Year	Europeans	Non-Europeans
1600	7193	7059
1614	11867	12364
1619	9706	13403
1636	10758	15046

The following procedure can now be implemented. We first optimize the model's sociodynamic parameters with relation to these demographic data. The relevant parameters—i.e. those parameters which play a role in how the numbers of L1 and L2 speakers in the model evolve—are listed in Table 10. The parameter estimates are found through numerical optimization using Differential Evolution, a derivative-free stochastic optimization method (Storn and Price 1997).[10] Briefly put, we start with an initial guess of the parameter values, run the model for a predefined time, and compare the numbers of L1 and L2 speakers predicted by the model against the empirical demographic numbers. The error between the model realization and the empirical data—i.e. the quantity the optimization algorithm seeks to minimize—is given by the objective function

$$\sqrt{\frac{1}{|T|}\sum_{t\in T}\big(O_{\mathrm{L1}}(t)-P_{\mathrm{L1}}(t)\big)^2+\frac{1}{|T|}\sum_{t\in T}\big(O_{\mathrm{L2}}(t)-P_{\mathrm{L2}}(t)\big)^2}.$$

Here, T denotes the set of time points in the data (i.e. the years for which demographic data is available), $O_{\mathrm{L1}}(t)$ denotes the empirical (observed) number of L1 speakers at time t, $P_{\mathrm{L1}}(t)$ denotes the modelled (predicted) number of L1 speakers at time t, and so on. In respect of the empirical demographic data, we take all Europeans to be L1 speakers and all non-Europeans to be L2 speakers (at least initially, i.e. at the time they are introduced into the population). This constitutes an idealization, of course, but in the absence of further empirical information it is difficult to see how one might proceed otherwise.

Table 10. Sociodynamic parameters to be optimized.

Parameter	Symbol
Birth rate	β
Expected proportion of L2 learners	$\mathrm{E}[\sigma]$
Length of immigration period	T_{L2}
Mean L2 learner age at immigration	m
S.D. of L2 learner age at immigration	s
Number of L1 speakers at time 0	N_0

[10] Specifically, adaptive, radius-limited DE with a "DE/rand/1/bin" mutation scheme was employed here (see Das and Suganthan 2011 for details).

Parameter	Symbol
Shape parameter of death distribution	k
Scale parameter of death distribution	λ
Population carrying capacity	K
Language contact start time	T_{L2}^{0}

The same procedure is then repeated with another guess of the parameter values, and the value of the objective function computed. The procedure is repeated until a given convergence criterion has been reached; here, the criterion used was to halt the optimization algorithm after a runtime of 24 hours.

Table 11. Sociodynamic parameters found through optimization (non-integers rounded to two-digit precision). Time is measured in units of years. Time 0 refers to year 1652 for Afrikaans (the year of settlement of the Cape by the Dutch) and year 1532 for Afro-Peruvian Spanish (the year of the arrival of the Spanish at the borders of the Inca Empire).

Parameter	Symbol	Afrikaans	APS
Birth rate	β	0.03	0.04
Expected proportion of L2 learners	$E[\sigma]$	0.53	0.69
Length of immigration period	T_{L2}	112	86
Mean L2 learner age at immigration	m	25.99	14.95
S.D. of L2 learner age at immigration	s	95.24	7.88
Number of L1 speakers at time 0	N_0	67	829
Shape parameter of death distribution	k	9.79	1.11
Scale parameter of death distribution	λ	228.03	230.3
Population carrying capacity	K	99006	26786
Language contact start time	T_{L2}^{0}	44	53

The optimized parameter values found using this procedure are listed in Table 11 for both Afrikaans and APS. Figure 9 illustrates the fit between model and data, for these optimized parameter values. We find that the optimized values are largely sensible, in the sense that the modelled trajectories correspond reasonably closely to the empirical data points. Some departures can be noted: for instance, in the case of Afrikaans, the empirical population size is consistently higher than the population size predicted by the model between the years 1700 and 1750; this is likely due to the fact that birth and migration rates need not be constant in real life, while in the model they are forced to remain constant over the entire simulation run for simplicity's sake. A second obvious departure from the model's prediction can be noted in APS: the number of Europeans in the year 1614 is quite a bit higher than the predicted number of L1 speakers, possibly for similar reasons.

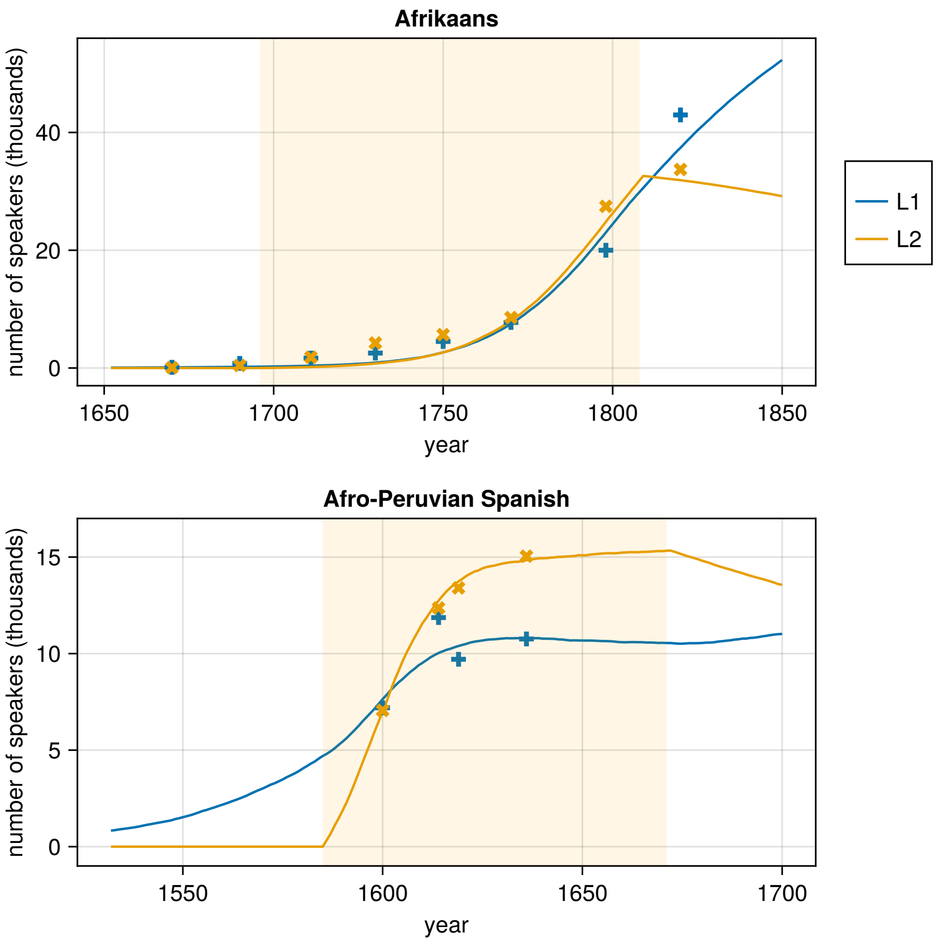

Figure 9. The lines give the numbers of L1 and L2 speakers predicted by the model, given the optimized sociodynamic model parameters in Table 11. The crosses are the empirical data (upright crosses: Europeans, slanted crosses: non-Europeans). The timespan highlighted in yellow indicates the period of L2 learning predicted by the model.

Next, the optimized sociodynamic parameter values are combined with linguistic parameters: a_1 and a_2, the advantages of the competing grammars, and d, the amount of L2-difficulty on grammar G_1. For the advantage parameters, estimates exist: as mentioned above, we have $a_1 = 0.7$ and $a_2 = 0.05$ in the case of APS by the method of Simonenko, Crabbé, and Prévost (2019). In the case of Afrikaans, Kauhanen (2022) takes $a_1 = a_2 = 1$ since there is no overlap between the forms produced by the two grammars. For the L2-difficulty parameter d, no estimates currently exist, and so we are forced to sweep over this parameter, observing how variation in d affects the model outcomes.

Figure 10 shows the resulting simulation outcomes for both case studies; in other words, plotted is the mean value of P, the weight on the L2-difficult grammar G_1, for each year from the beginning of the simulation up to present day. Each panel of the figure contains 10 curves, corresponding to 10 different (logarithmically-spaced) values of the L2-difficulty d.

For Afro-Peruvian Spanish, the simulation outcomes are consistent with both the empirical reality and the deterministic bifurcation threshold result from Kauhanen (2022): firstly, full simplification is never attained, no matter how large the value of d, and secondly, once L2 learning is removed, the representation of the L2-difficult grammar G_1 slowly creeps towards full usage, replicating the observation that younger speakers of APS are reverting back to using a consistent null subject grammar (Sessarego 2014).

For Afrikaans, the simulation results are roughly of the right shape, in the sense that once L2 learning is removed, the language does not begin to complexify again. (This is due to the fact that here, $a_1 = a_2$, and so neither grammar enjoys an advantage in the absence of external factors such as the presence of L2 learners.) However, the simulations fail to replicate the full simplification observed in the empirical Dutch-to-Afrikaans transition: the fully simplified state $P = 0$ is not reached even for very high L2-difficulties, such as $d = 100$. This appears to be because the length of the period of L2 learning (T_{L2}) found through the optimization is too short to advance the simplification to its extreme.

This last observation brings forth an important challenge in the sort of predictive modelling attempted here. The accuracy of any prediction must in general decrease the further into the future we look; attempting to model the status of a language in the year 2025 using a handful of demographic datapoints from 2–3 centuries ago is of course fraught with uncertainty. Although the above optimizations suggest relatively short periods of L2 learning for the two languages, this seems to contradict empirical reality. The case of Afrikaans is particularly poignant: according to Ethnologue (REF), the estimated number of L2 speakers of Afrikaans stood at 10.3 million in 2011, compared to only 7.2 million L1 speakers. Obviously, L2 learning of Afrikaans has never ceased; in fact, today there are more L2 speakers than L1 speakers of the language.

It would be tempting to attempt to address this issue by adding the contemporary L1 and L2 speaker estimates to the historical data and rerunning the optimization. Unfortunately, with current technology and implementation, simulating the model with millions (as opposed to thousands) of speakers is not computationally feasible

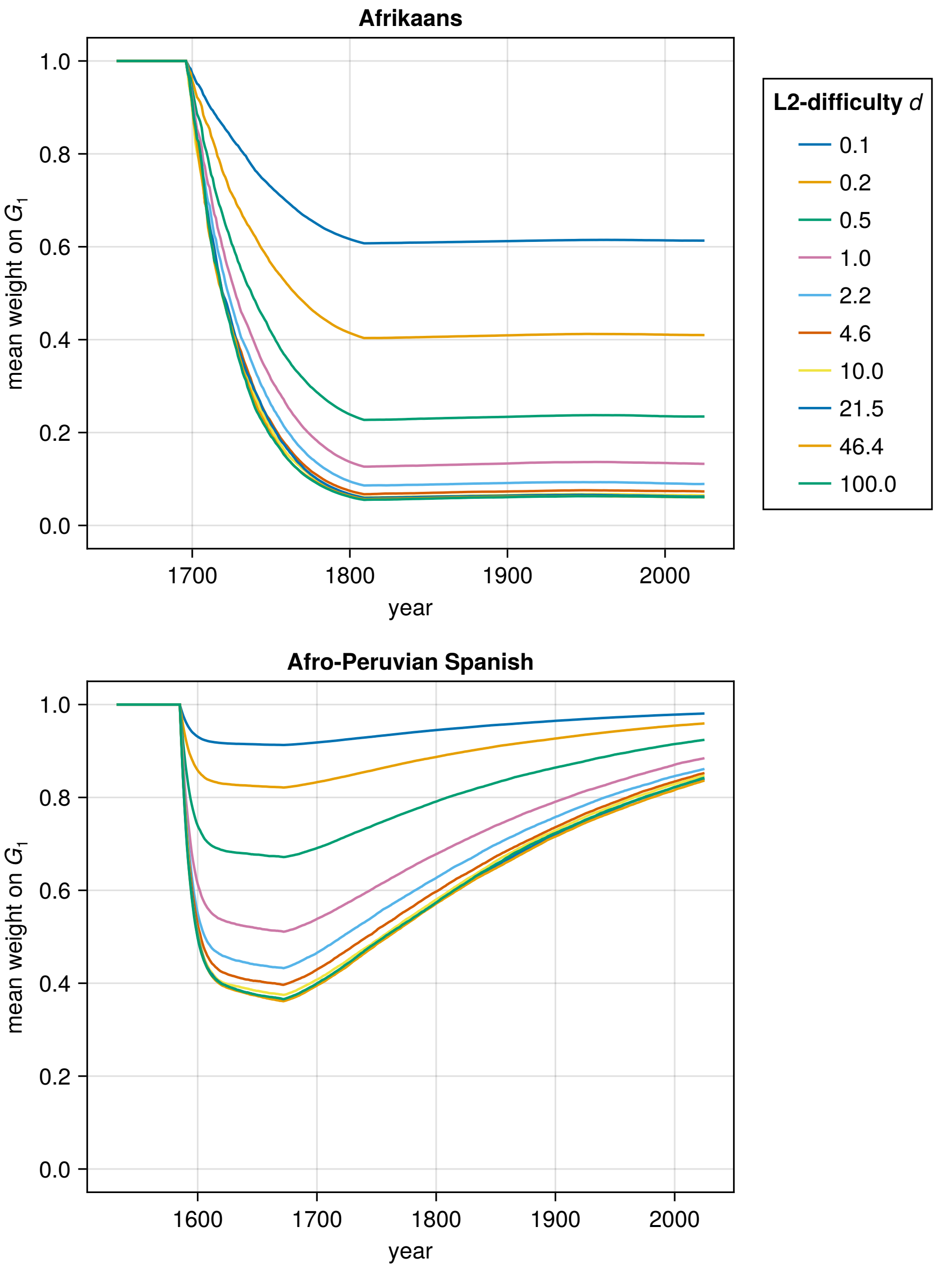

Figure 10. Simulations of Afrikaans and Afro-Peruvian Spanish assuming the optimized sociodynamic parameters in Table 11.

[FIXME: need to examine this! it might actually work, might just take, like, a month to run…] What we can relatively easily do, however, is to constrain the length of the L2 learning parameter (T_{L2}) in the optimization to have a high lower bound, such as 500 years (which guarantees that there will still be L2 speakers in 2025). Doing this yields the simulation outcomes visualized in Figure 11. From these results, we find that $P = 0$ is predicted to be an attracting state, albeit only for high values of L2-difficulty (over empirically realistic timescales). Figure 11 also shows the simulation outcome for APS with the assumption that L2 learning is never broken. The prediction of partial simplification remains, in line with the deterministic-limit results. However, the reversion back towards the L2-difficult grammar is now lost, as L2 learning is allowed to continue indefinitely.

The above simulations assumed a perfectly square lattice (aspect ratio $\rho = 1$). In light of the findings in Section 5.6, one may suspect that the lattice aspect ratio will also play a role in the diachronic dynamics. For this reason, a final simulation batch was carried out for Afrikaans, with the optimized sociodynamic parameters and assuming a lower bound of 500 years on the period of L2 learning. L2-difficulty d was varied from 0.1 to 10 and lattice aspect ratio ρ from 0.001 to 1000. Figure 12 visualizes the results as a heatmap. A region that favours full simplification exists. Although aspect ratio ρ plays a role, in the sense that very wide lattices ($\rho \gg 1$) inhibit full simplification, L2-difficulty plays the larger role in driving simplification.

5.8 Discussion

Here I discuss

Appendix

The purpose of this appendix is to prove that, given the simulation setup assumed above, the expected proportion of L2 learners in the population is completely determined by the birth and immigration rates β and τ:

$$\mathrm{E}[\sigma] = \frac{\tau}{\beta + \tau}.$$

More precisely, this is the asymptotic value of $\mathrm{E}[\sigma]$ if L2 learning is allowed to continue indefinitely. In practice, in the simulations conducted in this chapter, this number describes the peak proportion of L2 learners attained during the T_{L2} simulation steps when L2 learning occurs (cf. Figure 5).

Let X and Y refer to the numbers of L1 and L2 learners in the population, respectively, at some given simulation step. Moreover, let X' and Y' refer to these numbers at the immediately following simulation step. Let B refer to the number of L1 learners born at a given step, and let M refer to the number of L2 learners immigrated. With the assumptions made in the main text, we have $B \sim \mathrm{Bin}(\lfloor N(1 - N/K) \rfloor^+, \beta)$, where $N = X + Y$ is the total population size, K is a constant (the carrying capacity), and $\lfloor a \rfloor^+ = \lfloor a \rfloor$ if $a > 0$ and $\lfloor a \rfloor^+ = 0$ otherwise. Similarly, $M \sim \mathrm{Bin}(\lfloor N(1 - N/K) \rfloor^+, \tau)$.

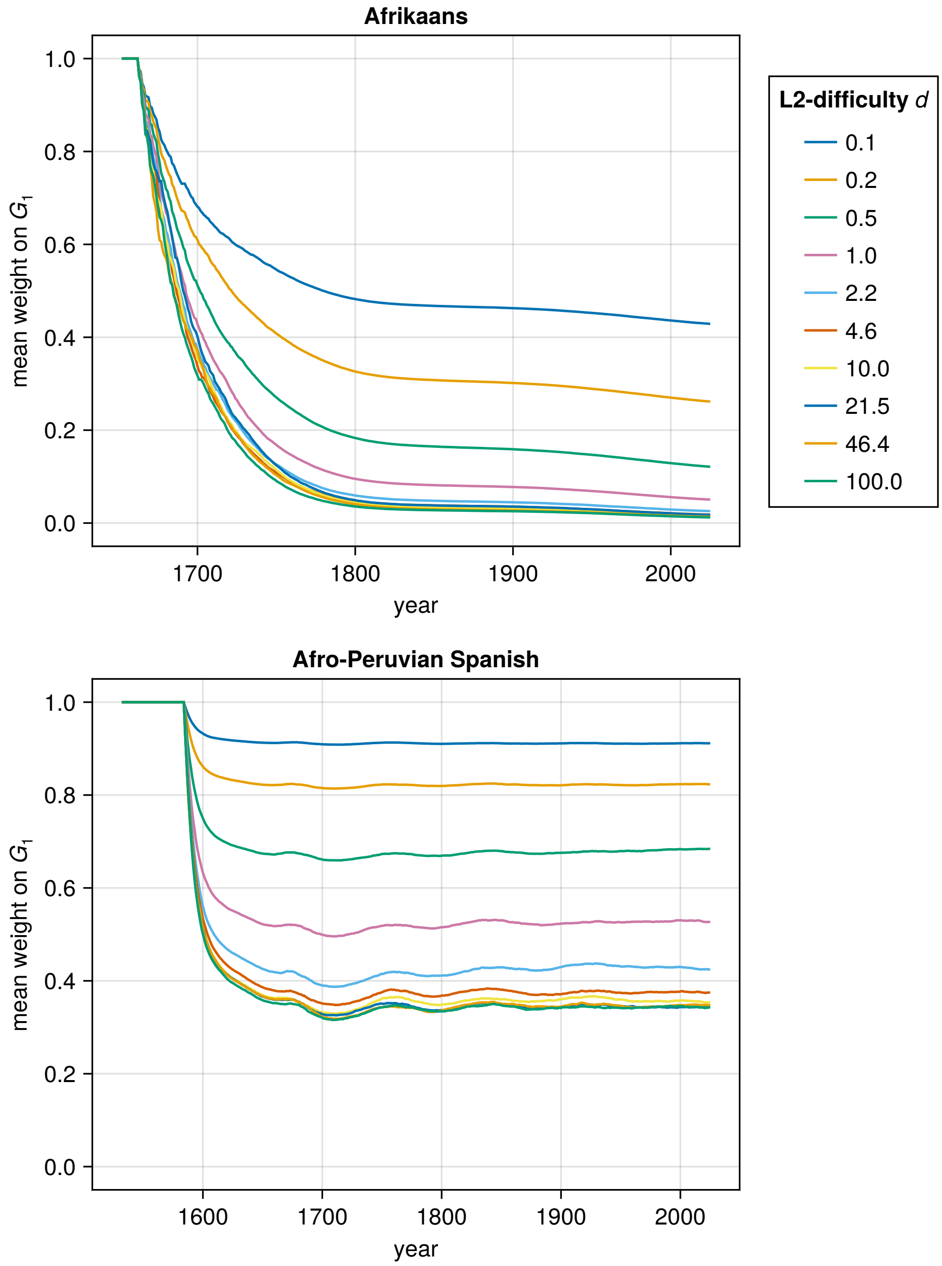

Figure 11. Simulations of Afrikaans and Afro-Peruvian Spanish using the optimized sociodynamic parameters, with the added constraint that the length of the period of L2 learning, T_{L2}, must be over 500 years. In effect, this constraint means that, over the timescales of interest here, the supply of L2 learners is never cut off.

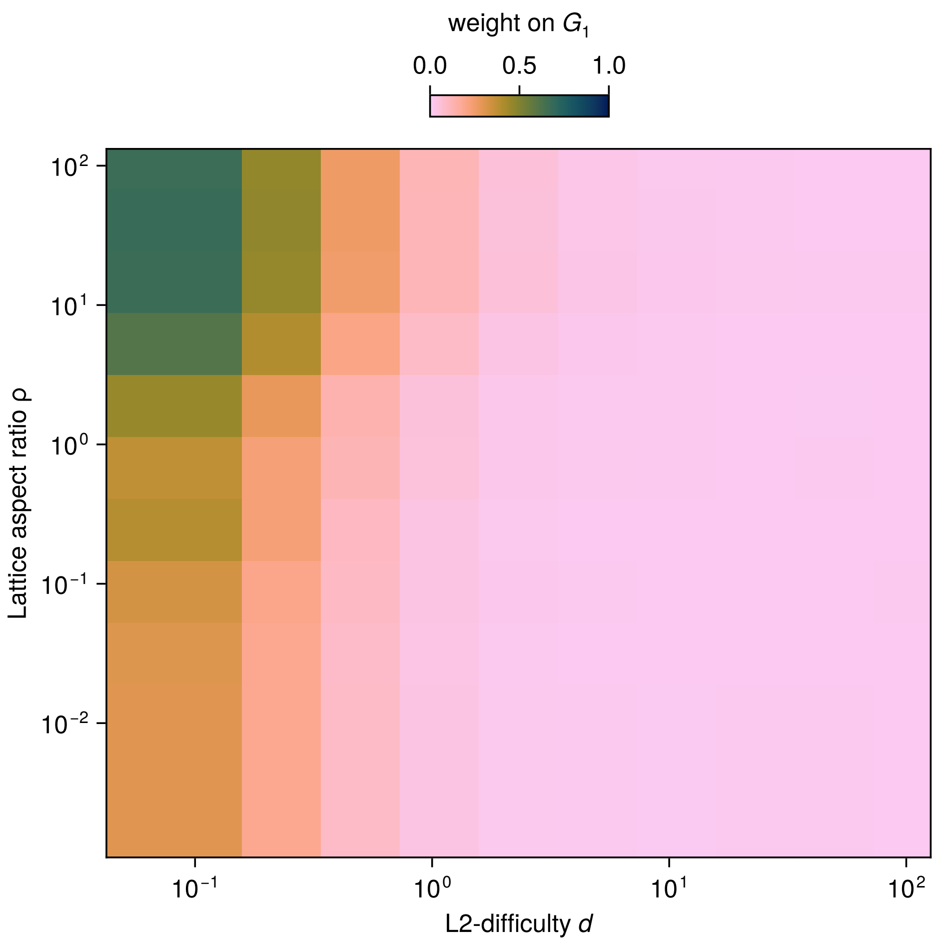

Figure 12. Mean weight on G_1 in year 2025 in Afrikaans, assuming the supply of L2 learners is never cut off, as a function of L2-difficulty d and lattice aspect ratio ρ. FIXME: this is a low-resolution version, will be replaced with a more detailed plot later

The expected numbers of L1 and L2 learners after one iteration are thus

$$\begin{cases} \mathrm{E}[X'] = \mathrm{E}[X] + \mathrm{E}[B] - \mu \mathrm{E}[X] \\ \mathrm{E}[Y'] = \mathrm{E}[Y] + \mathrm{E}[M] - \mu \mathrm{E}[Y] \end{cases}$$

where μ is a death rate (the exact value of this parameter will depend on the parameterization of the Weibull distribution assumed in the main text). The expected change between two consecutive steps is thus

$$\begin{cases} \mathrm{E}[X' - X] = \mathrm{E}[B] - \mu \mathrm{E}[X] \\ \mathrm{E}[Y' - Y] = \mathrm{E}[M] - \mu \mathrm{E}[Y] \end{cases}$$

Given that B and M are binomially distributed, the following holds:

$$\begin{cases} \mathrm{E}[X' - X] = \beta \lfloor N(1 - N/K) \rfloor^{+} - \mu \mathrm{E}[X] \\ \mathrm{E}[Y' - Y] = \tau \lfloor N(1 - N/K) \rfloor^{+} - \mu \mathrm{E}[Y] \end{cases}$$

Approximately, we have

$$\begin{cases} \mathrm{E}[X' - X] \approx \beta N(1 - N/K) - \mu \mathrm{E}[X] \\ \mathrm{E}[Y' - Y] \approx \tau N(1 - N/K) - \mu \mathrm{E}[Y] \end{cases}$$

At equilibrium, one has $\mathrm{E}[Y' - Y] = 0$ and so

$$\tau N(1 - N/K) = \mu \mathrm{E}[Y].$$

Taking expectations on both sides and dividing τ onto the right hand side,

$$\mathrm{E}[N] - \mathrm{E}[N]^2/K = \frac{\mu}{\tau} \mathrm{E}[Y].$$

Dividing both sides by $\mathrm{E}[N]$ we obtain

$$1 - \mathrm{E}[N]/K = \frac{\mu}{\tau} \mathrm{E}[\sigma],$$

from which

$$\mathrm{E}[\sigma] = \frac{\tau}{\mu}(1 - \mathrm{E}[N]/K). \tag{5}$$

Carrying out the same exercise starting from $\mathrm{E}[X' - X] = 0$, one obtains

$$\mathrm{E}[\sigma] = 1 - \frac{\beta}{\mu}(1 - \mathrm{E}[N]/K). \tag{6}$$

Equating Equation 5 and Equation 6 allows us to solve for $\mathrm{E}[N] = \mathrm{E}[X + Y]$, the expected total population size at equilibrium:

$$\mathrm{E}[N] = K\left(1 - \frac{\mu}{\beta + \tau}\right).$$

Substituting this back to Equation 5 finally gives us

$$\mathrm{E}[\sigma] = \frac{\tau}{\mu}\left(1 - \left(1 - \frac{\mu}{\beta+\tau}\right)\right) = \frac{\tau}{\mu}\cdot\frac{\mu}{\beta+\tau} = \frac{\tau}{\beta+\tau}$$

as desired.

Thus, in order to obtain a desired L2 learner proportion σ given a fixed birth rate β, we need to select the immigration rate τ such that

$$\tau = \frac{\sigma}{1-\sigma}\beta.$$

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